

Ahmed Hagraas

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SUMMARY

Software Engineer with hands-on experience in full-stack development and AI-enabled systems, combining strong software engineering fundamentals with practical delivery in business and research environments. Experienced in building web platforms, backend APIs, workflow automation tools, and machine learning solutions across project operations, smart applications, and aquaculture. Comfortable working in Agile teams and focused on clean architecture, maintainable code, and measurable product impact.

EXPERIENCE

Software Developer Intern

Global Project Services (GPS)

Jun 2024 – Aug 2024, Egypt

- Developed a full-stack project management platform using React.js and Node.js to manage and monitor 40+ active modular construction projects, replacing spreadsheet-based workflows.
- Designed and implemented a PostgreSQL database schema with role-based access control (RBAC) and JWT authentication to support secure internal operations.
- Integrated an AI-powered assistant to streamline proposal drafting and improve client interaction, helping automate repetitive business processes.
- Built a quotation automation tool that reduced document preparation time by approximately 40–45%, improving speed and consistency across project teams.
- Developed and maintained RESTful APIs to handle project data, user roles, quotation workflows, and system interactions.
- Collaborated within an Agile/Scrum environment through stand-ups, sprint planning, testing, and iterative feedback with technical and business stakeholders.

Machine Learning Intern

Bright Network

Sep 2024 – Oct 2024, UK

- Built automated data preparation pipelines using SQL and Python to clean, transform, and structure datasets for predictive modelling tasks.
- Applied PyTorch and Scikit-learn across multiple technical case studies focused on classification, evaluation, and model selection.
- Produced model performance summaries using precision, recall, F1-score, and ROC-based analysis to support technical decision-making.
- Worked in an Agile team setting and documented findings clearly to ensure reproducibility and strong handover quality.

PROJECT

Future Fridges Mobile Application

Team Project

- Collaborated in an Agile team to develop a smart fridge application focused on food inventory visibility, expiry tracking, and day-to-day household usability.
- Contributed to interface development and system integration, linking UI flows with application logic to support dynamic user interactions.
- Supported design planning and prototyping using Figma to improve usability and produce a clearer end-to-end user journey.
- Participated in testing activities including validation, user testing, and acceptance testing, helping document findings and improve reliability before final delivery.

Aquaculture Food Management & Waste Reduction System

Final-Year Project Contribution

- Led the food management and waste reduction side of an aquaculture AI project aimed at improving feeding efficiency and reducing operational loss in fish farms.
- Worked on feeding logic and management insights that supported better portion control, waste monitoring, and more sustainable aquaculture practices.
- Collaborated with the wider project team alongside computer vision and deep learning components used to monitor fish behaviour and feeding activity.
- Helped shape a practical system objective that connected technical outputs with real farm efficiency, sustainability, and cost-reduction goals.

Developer Salary Prediction

- Built a machine learning model using the Stack Overflow developer survey dataset to predict salary bands from demographic and technical features.
- Performed exploratory data analysis, preprocessing, feature engineering, and model evaluation using precision/recall and ROC-based metrics.

Event Booking Web System

- Developed a full-stack event booking web application with authentication, structured backend services, and real-time event filtering.
- Designed data persistence and API architecture to support reliable booking flows, maintainability, and clear error handling.

EDUCATION

BSc (Hons) Software Engineering, Upper Second-Class Honours (2:1)

Nottingham Trent University • Nottingham, UK • 2022 – 2025

- Key Modules: Advanced Software Engineering, AI & Machine Learning, Cloud Computing, Database Systems, Mobile Application Development, Internet Application Programming.
 - Course Representative for final-year Software Engineering.
 - Contributed to Formula Student activities, supporting data-led analysis and collaborative team delivery.
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SKILLS

Python, JavaScript, TypeScript, C#, C++, SQL, Java

React.js, Node.js, ASP.NET Core, REST APIs, HTML, CSS

PyTorch, Scikit-learn, CNNs, YOLO, EDA, data preprocessing, model evaluation

PostgreSQL, SQL Server, MySQL, MongoDB

Git, Docker, Jenkins, Jupyter Notebook, Figma

OOP, Data Structures & Algorithms, RBAC, JWT, Agile/Scrum, SDLC
